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Factors associated with medical students' speaking-up about medical errors: A cross-sectional study

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ABSTRACT

Background: Training medical students to speak up when they witness a potential error is an important competency for patient safety, but details regarding the barriers that prevent medical students from effectively communicating are lacking. Therefore, this study aimed at exploring the factors affecting medical students' willingness to speak up for patient safety when a medical error was observed.

Methods: This is a cross-sectional study at a medical university in Taiwan, and 151 medical students in clinical clerkship completed a survey including demographic characteristics, conflict of interests/social relationship, personal capability, and personality and characteristics of senior staff domains. Data were analyzed using *t*-test.

Results: Three of five items in the conflict of interests/social relationship domain showed statistically significant importance, including 'I am afraid of being punished' (Mean difference, MD = 0.37; $p < 0.01$), 'I do not want to break unspoken rules' (MD = 0.55; $p < 0.01$), and 'I do not want to have bad team relationship' (MD = 0.58; $p < 0.01$). Two items (perception of knowledge/understanding and communication skills) in the personal capability domain were significantly important to speaking up. Six of 10 items in personality and characteristics of senior staff domain were rated significantly important in deciding to speak up. The top three factors of them were senior personnel with 'Grumpy' personality (MD = 1.20; $p < 0.01$), 'hierarchy gap' (MD = 1.12; $p < 0.01$), and senior personnel with 'Stubborn' personality (MD = 1.06; $p < 0.01$).

Conclusion: Our findings demonstrated medical students' perspectives on barriers to speaking up in the event of medical error. Some factors related to characteristics of senior staff could compromise medical students' ability to speak up in the event of medical error. These results might be important for medical educators in designing personalized educational activities related to medical students' ability to speak up for patient safety.

KEYWORDS
General; curriculum; undergraduate

Introduction

Patient safety is an important issue in medical care, and speaking up when a medical error is about to occur is an essential but difficult task in the prevention of adverse outcomes (Okuyama et al. 2014). Moreover, the timely response to medical errors by healthcare team staff could effectively prevent the occurrence of future medical errors (Belyansky et al. 2011). Evidence shows that ineffective communication between health professionals, especially between those who provide most of the clinical care and those in leadership positions is common, and has a negative impact on patient safety (Greenberg et al. 2007). Hesitancy to speak up may be an important contributing factor to communication error (Okuyama et al. 2014; Green et al. 2017). Therefore, it is a health priority to explore the human factors involved in speaking up when medical errors occur, and to identify how effective educational interventions can improve a patient care team member's willingness to speak up to improve patient safety in a clinical setting, that can then be applied in the design of education for effective medical communication.

The most common barriers that impede healthcare professionals' speaking up to medical errors are fear of

Practice points

- Medical students' intention to speak up in the event of a medical error could be affected by factors regarding conflict of interests/social relationships.
- Medical students' intention to speak up in the event of a medical error could be affected by some factors regarding personal capability and personality.
- Medical students' intention to speak up in the event of a medical error could be affected by factors regarding the characteristics of senior staff.
- Female students care about the age gap between them and seniors more than male students.

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retribution, professional hierarchy, personal capability, interpersonal factors, and lack of training (Omura et al. 2017). Professional hierarchy is defined as the often one-sided direction of command from senior doctors to emergency medical technicians, nurses, and younger doctors. This hierarchy between senior doctors and junior doctors and doctors and nurses impairs performance by discouraging lower-ranking team members from voicing their potentially valuable perspectives and insights (Anicich et al. 2015). The inability of lower-ranking team members to voice their response to potential medical errors has been observed within multiple hospital specialties. For example, in a surgery department, surgical errors were under-reported in a discouraging environment, where even trainees witnessed the errors of the senior surgeon (Salazar et al. 2014). A recent nursing study reported that a hospital's hierarchical system is not conducive to reporting medical errors, and hospital leadership commitment often has no plans to improve its reporting of medical errors. Despite nurses' experience with, and importance to clinical practice, they are usually unaware of the direct impact hospital culture has on patient safety (Levine et al. 2020). It is essential that these healthcare providers understand this connection, because senior healthcare providers, including senior physicians and nursing professionals, play an important role in encouraging a team communication atmosphere and can improve patient safety (Salazar et al. 2014; Levine et al. 2020). Studies highlight that a flat hierarchical gradient between healthcare team members enabled effective communication and better patient care (Green et al. 2017). The associated factors, including gender, power relationship, and leadership may also impact the health staff's willingness to speak up for their doubts of potential medical errors. There is evidence of the effectiveness of an appropriate education program to improve the willingness of health professionals to speak up, yet the effect in different cultural contexts remains to be explored (Salazar et al. 2014; Pattni et al. 2017).

Cultural factors in different countries may influence the effectiveness of assertiveness communication training in health care systems (Omura et al. 2017). In particular, there are large differences in hierarchy and education health care systems between Western and Eastern cultures. For example, the hierarchy of power is more established in Eastern countries than in the Western countries, cultural dimensions are also important to clinical training and practice (Meeuwesen et al. 2009; Morrow et al. 2013), and these differences can create significantly different models of education including those for medical communication. In Western culture, an open education that encourages students to speak up is promoted, while a 'spoon-fed,' didactic education is more common in the eastern culture. Multifaceted factors to challenging medical errors in residents have been revealed in previous studies (Beament and Mercer 2016; Guris et al. 2019), but few have explored the associated factors in the medical students, a population at one of the lowest ranks of the health care system (McGugan 2002; Caldicott and Faber-Langendoen 2005; Schwappach and Niederhauser 2019).

Medical students are often the first line of health professionals who are expected to transfer important clinical information to senior physicians and play a vital role in preventing potential medical errors and improving patient safety. Current studies found the training of speaking up among medical students important to patient safety, but details regarding the barriers that prevent medical students from effectively communicating are lacking (McGugan 2002; Caldicott and Faber-Langendoen 2005; Schwappach and Niederhauser 2019). Therefore, this study aimed at identifying and investigating the factors affecting medical students' willingness to speak up for patient safety when a medical error was observed.

Methods

Study design and participants

The authors conducted a cross-sectional study at a simulation training center of a medical university in Northern Taiwan from April to November 2018. In Taiwan, medical students are required to complete seven years of medical school that includes the final two years in with clinical clerkship in the 6th year and internship during the 7th year. Inclusion criteria for medical students were: (1) undertaking their clinical clerkship training, (2) enrolled in a required patient safety training course. Because this study mainly aimed to describe factors of speaking up in the event of medical errors among medical students. Simple statistical techniques were used. The minimum requirement of cases was 128 according to the prior analysis of sample size with alpha 0.05 (type I error probability), power 0.95 (1 – type II error probability), and Cohen's d 0.5 (medium effect size) for independent *t*-test using G*Power 3.1 for Microsoft Windows (Appendix Figure). A total of 151 medical students received survey. Before students completed the survey, we had a formal instruction session to inform them that their responses were anonymous, would be helpful for understand their perspectives, and would not affect their grades or their clinical training. To avoid hierarchy pressure, the lead instructor of this course did not attend any part of the survey process, and they were encouraged to complete the survey comfortably without senior visiting staff supervision. All the medical students completed the survey. Thus, this sample size reached the minimum requirement of cases for our exploratory study. The study was approved by the institutional review board of the medical university.

Questionnaire design and development

The authors developed a questionnaire based on prior studies (Kobayashi et al. 2006; Schwappach and Niederhauser 2019; Peadon et al. 2020). An expert panel consisting of physicians, nurses, and medical education faculty members reviewed the questionnaire. The 21-item questionnaire consists of demographic characteristics, and conflict of interests/social relationship, personal capability, and personality and characteristics of senior staff domains. Participants were asked to respond to Conflict of interests/social relationship domain questions using a 5-point Likert scale ranging from 'not important' (point 1) to 'very important' (point 5) based on the following clinical scenario: 'As part of a clinical medical team in an acute care setting, you observe a potential medical error initiated by the senior personnel that requires you to question that senior personnel to prevent the medical error from happening. What would be the considerations that could affect your response?' The conflict of interests/social relationship

Table 1. Demographic characteristic and speaking-up experience in medical students ($N = 151$).

Variable	Male ($n = 95$)	Female ($n = 56$)	Total ($n = 151$)
Age (years), mean (range)	23.5 (21–32)	24.1 (22–33)	23.8 (21–33)
Experience of speaking up in the event of medical error			
'No,' 52 (%) of students	47 (49)	31 (55)	78 (52)
'Yes,' 48 (%) of students	48 (51)	25 (45)	73 (48)
Had experience of being blamed after speaking up in the event of medical error ($n = 73$)			
'No,' 86 (%) of students	43 (90)	20 (80)	63 (86)
'Yes,' 14 (%) of students	5 (10)	5 (20)	10 (14)

domain included five items regarding cultural factors (penalty/reward), environmental factors (workplace atmosphere, acceptance or rejection), and interpersonal factors (self-image, value, and pride) (Kobayashi et al. 2006). The personal capability domain included two items; students' self-rating of their perception of knowledge and communication skills that would affect their ability to speak up. The personality and characteristics of the senior staff domain contain 10-items that can compromise speaking up behavior (Schwappach and Niederhauser 2019; Peadon et al. 2020). We assessed the personality and characteristics of senior staff domain using a 6-point Likert scale (completely agree, agree, slightly agree, slightly disagree, disagree, completely disagree), and included items of hierarchy gap, age gap, personality, gender, and different professionalism (Appendix Table). The survey also collected demographic characteristics including gender, age, and experiences of speaking-up in the event of medical errors.

Data analysis

We analyzed data using SPSS version 19.0 (IBM Corp., Armonk, NY, USA) for Microsoft windows. We report descriptive statistics for age, experiences of speaking up or being blamed after speaking up. Age is presented in mean with range (minimum to maximum), while experiences of speak-up and being blamed are shown in the number of events. We stratified data by gender because gender difference has been recognized as an important confounder in the literature (Martinez et al. 2017; Roussin et al. 2018). We conducted an independent-sample *t*-test to compare the mean and standard deviation of each item in conflict of interests/social relationship domain, personal capability domain, and personality and characteristics of senior staff domain, respectively, among male and female medical students, and participants who did or did not report an experience of speaking-up or being blamed for an error. We also calculated the correlations between age and potential factors of remaining silent in observing medical errors using Pearson correlation. To examine how one's ability to speak up was affected by conflict of interests/social relationship factor and personal capability factors, a critical threshold value of 3.5 for conflict of interests/social relationship domain, and the critical threshold value of 4.0 for the senior personal domain were used for one-sample *t*-test. To clarify patterns of factors of speaking-up in the event of medical errors within different major characteristics and experiences, the subgroup of sex and experience of speaking-up were carried out. Any result with *p*-values <0.05 was considered statistically significant.

Results

One hundred fifty-one medical students with a mean age of 23.8 years (ranged from 21 to 33 years) completed the study, including 95 male students and 56 female students (Table 1). Seventy-three students (48%) reported experiencing a medical error committed by their senior staff. For those students who reported speaking up experience, 10 students (14%) were blamed by their senior faculty (who made the mistake) for making them look bad in front of others after speaking up about a medical error.

Three items in conflict of interests/social relationship domain, two items in personal capability domain, and six items in personality and characteristics of senior staff domain were identified as main factors of refusing to speak up among medical students (Table 2). The three items in the conflict of interests/social relationship domain that showed statistical significant importance were, 'I am afraid of being punished' (Mean difference, MD $= 0.37$; $p < 0.01$), 'I do not want to break unspoken rules' (MD $= 0.55$; $p < 0.01$), and 'I do not want to have bad team relationship' (MD $= 0.58$; $p < 0.01$). Similar patterns were observed in medical students who had and had no experience with speaking up. Regardless of whether medical students had positive or negative experiences after speaking up for a medical error event, their reports also indicate these three items are important in deciding to speak up. In the personal capability domain, students reported that 'My professional knowledge' (MD $= 0.89$; $p < 0.01$) and 'My communication skill' (MD $= 0.61$; $p < 0.01$) were important in making the decision to speak up. Interestingly, it appears there was no difference in any of the items between those blamed and not blamed after speaking up ($p > 0.05$).

Among all participants, six significant items in the personality and characteristics of senior staff domain were 'hierarchy gap' (MD $= 1.12$; $p < 0.01$), 'age gap' (MD $= 0.27$; $p = 0.01$), 'Strict' (MD $= 1.01$; $p < 0.01$), 'Grumpy – senior personnel has unstable emotion and is easily angers' (MD $= 1.20$; $p < 0.01$), 'Stubborn – senior personnel would insist and would not change his/her thoughts even I speak up' (MD $= 1.06$; $p < 0.01$), and 'interest concern' (MD $= 0.32$; $p < 0.01$). There were no obvious gender differences in 5 of these items. There is no level of significance for the item 'age gap' (MD $= 0.02$; $p = 0.88$) in male students, but female students reported a significant score in this item (MD $= 0.69$; $p < 0.01$). Males had significantly higher score than critical value on item 'Being blamed or penalized' (MD $= 0.45$; $p < 0.01$) in conflict of interests/social relationship domain and item 'Interest conflict' (MD $= 0.36$; $p = 0.01$) in senior domain while there is no significance for female students.

The results show no statistically significant differences between age, gender, and experiences in the conflict of interests/social relationship and personal domains, and only one significant difference in the personality and characteristics of senior staff domain between genders (Table 3). Females (4.70 ± 1.03) had significantly higher scores on the age gap than males (4.02 ± 1.38; $p < 0.01$). There was not a significant correlation for any of the items with the age of the students in our study.

Table 2. One-sample *t*-test for conflict of interests/social relationship and personal capability factors (critical value = 3.5), and senior personnel factors (critical value = 4).

					Experience of speaking up				Experience being blamed for speaking up					
	Male		Female		Yes		No		Yes		No		Overall	
Item	MD	*p*	MD	*p*	MD	*p*	MD	*P*	MD	*p*	MD	*p*	MD	*p*
Conflict of interests/social relationship														
1. Penalty/repercussion	0.45	0.00*	0.24	0.07+	0.44	0.00*	0.32	0.00*	0.80	0.00*	0.39	0.00*	0.37	0.00*
2. Reward or prestige	−0.33	0.00*	−0.54	0.00*	−0.47	0.00*	−0.37	0.00*	0.00	1.00	−0.55	0.00*	−0.41	0.00*
3. Social acceptance or rejection	0.62	0.00*	0.45	0.00*	0.63	0.00*	0.49	0.00*	0.70	0.01*	0.61	0.00*	0.55	0.00*
4. Teamwork/Professionalism	0.03	0.68	0.03	0.82	0.13	0.23	−0.08	0.42	0.30	0.33	0.10	0.39	0.03	0.64
5. Relationship between senior and me	0.50	0.00*	0.71	0.00*	0.63	0.00*	0.54	0.00*	0.90	0.00*	0.58	0.00*	0.58	0.00*
Perceived personal capability domain														
6. Knowledge/understanding	0.86	0.00*	0.94	0.00*	0.88	0.00*	0.90	0.00*	0.70	0.06+	0.90	0.00*	0.89	0.00*
7. Communication skills	0.54	0.00*	0.73	0.00*	0.49	0.00*	0.72	0.00*	0.20	0.57	0.53	0.00*	0.61	0.00*
Personality and characteristics of senior staff domain														
8. Hierarchy gap	1.05	0.00*	1.23	0.00*	1.25	0.00*	1.01	0.00*	1.10	0.00*	1.27	0.00*	1.12	0.00*
9. Age gap	0.02	0.88	0.69	0.00*	0.36	0.03*	0.21	0.15	0.40	0.40	0.35	0.05*	0.27	0.01*
10. Strict	0.93	0.00*	1.14	0.00*	1.01	0.00*	1.01	0.00*	0.70	0.17	1.06	0.00*	1.01	0.00*
11. Grumpy	1.13	0.00*	1.31	0.00*	1.21	0.00*	1.19	0.00*	0.90	0.09	1.25	0.00*	1.20	0.00*
12. Stubborn	1.04	0.00*	1.09	0.00*	1.15	0.00*	0.97	0.00*	1.00	0.01*	1.18	0.00*	1.06	0.00*
13. Male gender	−1.02	0.00*	−0.91	0.00*	−0.94	0.00*	−0.96	0.00*	−0.50	0.41	−1.02	0.00*	−0.97	0.00*
14. Female gender	−1.31	0.00*	−1.30	0.00*	−1.35	0.00*	−1.23	0.00*	−1.10	0.02*	−1.39	0.00*	−1.30	0.00*
15. Interest conflict	0.36	0.01*	0.26	0.13	0.31	0.05*	0.35	0.02*	0.10	0.85	0.34	0.04	0.32	0.00*
16. Doctor background	−0.24	0.09	0.17	0.37	−0.07	0.69	−0.09	0.57	0.00	1.00	−0.08	0.66	−0.08	0.46
17. Nurse background	−0.80	0.00*	−0.71	0.00*	−0.77	0.00*	−0.73	0.00*	−1.30	0.00*	−0.68	0.00*	−0.76	0.00*

MD: mean deviation.
Note. $N = 153$.
$^*p < 0.05$, two-tailed.

Table 3. Characteristics on conflict of interests/social relationship domain, personal capability and personality and characteristics of senior staff domains.

						Speaking up					Consequences of speaking up					
	Male		Female			Experience		No experience			Blamed		Not blamed			Age
Item	*M*	*SD*	*M*	*SD*	*p*	*M*	*SD*	*M*	*SD*	*p*	*M*	*SD*	*M*	*SD*	*p*	(*r*)
Conflict of interests/social relationship domain																
1	3.95	0.77	3.74	0.97	0.16	3.94	0.80	3.82	0.88	0.37	4.30	0.48	3.89	0.83	0.13	0.00
2	3.17	1.01	2.96	1.10	0.25	3.03	1.10	3.13	0.99	0.56	3.50	1.08	2.95	1.09	0.14	0.04
3	4.12	0.73	3.95	0.79	0.18	4.13	0.79	3.99	0.73	0.27	4.20	0.63	4.11	0.81	0.75	−0.00
4	3.54	0.87	3.53	0.87	0.94	3.63	0.88	3.42	0.85	0.15	3.80	0.92	3.60	0.88	0.50	0.11
5	4.00	0.67	4.21	0.71	0.06	4.13	0.70	4.04	0.69	0.44	4.40	0.52	4.08	0.71	0.18	−0.05
Personal capability domain																
6	4.36	0.60	4.44	0.80	0.48	4.38	0.70	4.40	0.67	0.84	4.20	1.03	4.40	0.64	0.56	−0.07
7	4.03	0.91	4.23	0.80	0.21	3.99	0.88	4.22	0.86	0.11	3.70	1.06	4.03	0.85	0.27	0.03
Personality and characteristics of senior staff domain																
8	5.05	1.21	5.22	0.73	0.33	5.25	1.09	5.01	1.01	0.17	5.10	0.74	5.27	1.14	0.65	−0.06
9	4.02	1.38	4.70	1.03	0.00*	4.36	1.36	4.21	1.24	0.48	4.40	1.43	4.35	1.36	0.91	0.01
10	4.93	1.18	5.14	0.96	0.25	5.01	1.18	5.01	1.03	1.00	4.70	1.49	5.06	1.13	0.37	−0.06
11	5.13	1.11	5.31	1.06	0.32	5.21	1.09	5.19	1.11	0.94	4.90	1.52	5.25	1.02	0.34	0.02
12	5.04	1.02	5.09	1.14	0.81	5.15	0.94	4.97	1.17	0.31	5.00	1.05	5.18	0.93	0.59	−0.01
13	2.98	1.50	3.12	1.51	0.57	3.06	1.50	3.04	1.51	0.95	3.50	1.84	2.98	1.44	0.32	−0.14
14	2.69	1.22	2.72	1.19	0.89	2.65	1.22	2.77	1.19	0.57	2.90	1.20	2.61	1.23	0.50	−0.07
15	4.36	1.30	4.26	1.26	0.63	4.31	1.33	4.35	1.26	0.85	4.10	1.60	4.34	1.29	0.60	0.12
16	3.76	1.37	4.17	1.45	0.07	3.93	1.44	3.91	1.39	0.93	4.00	1.70	3.92	1.41	0.87	−0.08
17	3.20	1.26	3.29	1.24	0.66	3.23	1.25	3.27	1.26	0.86	2.70	1.06	3.32	1.27	0.15	0.06

M: mean; SD: standard deviation.
$^*p < 0.05$, two-tailed.

Discussion

This descriptive cross-sectional study provides evidence as to why medical students do or do not speak up when medical errors occur. We identified factors that could influence medical students' decision to speak up in the event of medical error including penalties/repercussions, social acceptance or rejection, relationship between senior and me, perception of knowledge/understanding, perception of communication skills, hierarchy gap, age gap, a certain type of personality related to seniors such as strict, grumpy, and stubborn, and interest conflict. We identified that female students care about the age gap between them and seniors more than male students. There are no perspectives differences among medical students between groups with speaking up experiences and without speaking up experiences.

Potential penalties/repercussions as a consequence of speaking up were one of the main factors affecting a medical student's decision to keep silent when noticing a medical error initiated by their superiors. In contrast, the potential reward does not encourage the medical student to speak up. These findings are similar when comparing the perspectives of residents in United States. About 74.2% of them report penalty/repercussion is an important and very important factor, and reward/prestige is not a clear factor affecting them to speak up in the event of medical error. Scores of reward/prestige reported by residents in Japanese were significantly higher than the scores reported by residents in the US (Kobayashi et al. 2006). This indicates that regional and cultural differences should be considered when discussing potential barriers in speaking up in the event of medical error. Although Taiwan and Japan are often considered more similar in terms of their regional and culture, the factors

impeding our medical students from speaking up when encountering medical errors are more similar to the resident in the United States.

Among health care professionals, it is known that influencing factors of speaking up to prevent medical error include motivation (Lyndon 2008; Lyndon et al. 2012), hospital administrative support (Simpson and Lyndon 2009; Churchman and Doherty 2010), individual factors (Tangirala and Ramanujam 2008; Churchman and Doherty 2010), the perceived efficacy of speaking up (Tangirala and Ramanujam 2008; Rutherford et al. 2012), and the perceived safety of speaking up (Kobayashi et al. 2006; Attree 2007; Okuyama et al. 2014). Health care providers in Taiwan have concerns about speaking up to prevent or report the medical error (JCT 2019, 2020). Few studies focus on medical students, the lowest-ranking health care team members. In another study, physicians and nurses were fearful of being blamed, accused, or branded after making a mistake because of the potential consequence of losing their status, jobs, and positions as well as uncertainty and legal sanctions (Soydemir et al. 2017). Educational interventions targeting medical personnel's concerns over speaking up are warranted. Clinical mentors could play an important role for medical students to discuss and address some of their concerns and help them to develop their patient safety attitude.

Gender difference may be one of the important factors underlying one's ability to speak up, but few studies reported the gender difference in the behavior to challenge the senior practitioners among medical students. In this study, we identified the age gap between senior personnel and students to be a significant influential factor for not speaking up in the event of a medical error for female medical students compared to male medical students. We also found that female medical students tend to care about the teamwork/professionalism/hierarchy between themselves and senior personnel more than male medical students. Previous work suggests that consideration of gender differences in superiors is important for effective teaching and assessment, and gender plays an important role in speaking out for a medical error (Pattni et al. 2017). In our study context, however, we found that the gender of superiors is not a strong factor for our students' speaking up behavior. A recent study also indicates that a male-biased medical hierarchy could compromise female nurses, who are considered lower power in that context, to voice their concerns (Perry et al. 2018). Educators should be aware that different genders have different perspectives that could impact students' learning and clinical performance. Gender differences should be considered when learner-centered educational activities are developed for medical students.

To encourage speaking up in the event of medical errors, hospital policies and support measures are strongly needed for medical personnel and students to address their concerns. This includes the workplace's acceptance or rejection, teamwork and relationship would affect medical students' decision of speaking up. Professional, national, and organizational cultures could also affect individual attitudes (Helmreich and Merritt 2019). Administrators and clinical practitioners should work together to build up a positive working atmosphere to facilitate opinion exchange and effective dialogue between care team members about what is best for the patient instead of rejection of different opinions or thoughts.

Our results show that students' perception of knowledge/understanding and communication skills act as important factors in their decision to transfer information to a senior physician. Similar findings were reported by Kobayashi and Pian-Smith's study of residents in the United States and Japan (Kobayashi et al. 2006). Thus, developing a curriculum using clinical scenarios that highlight professional knowledge and communication training can be helpful in fostering speaking up behavior in medical students. Simulation scenarios and role play are feasible training strategies for students to practice and develop their communication skills in difficult situations. Also, we noted that having previous experience when speaking up about a medical error does not necessarily influence students' perspective and consideration of challenging superiors. This indicates that gaining experience alone may not change their perspective. In our survey, some students reported negative experiences of being blamed or punished after speaking up in the event of medical error that could even compromise the behavior of speak up in the future. A mindful support system to assist and mentor these students who encounter this kind of negative experience may be helpful.

Studies by Okuyama et al. showed that speaking up for patient safety was associated with the attitudes of leaders (Okuyama et al. 2014). In our study, we further identified certain types of personality of senior staff, such as strict, grumpy, and stubborn could inhibit medical students' willingness to speak up in the event of medical error. Educational activities that increase the senior faculty's awareness of personality characteristics and their impact on effective communication may be useful in building up a mentally safe environment to encourage speaking up to maintain patient safety.

Moreover, the most recent survey study on this issue only targeted six aspects of barriers to speaking-up including (a) unclear risk, (b) fear negative reaction, (c) presence of patients, (d) uncertainty about how to, (e) ineffectiveness, and (f) reaction not predictable, yet it recruited more participants ($n = 326$) (Schwappach and Niederhauser 2019). We conducted a more in-depth survey with conflict of interests/social relationship factors, personal factors, and authority characteristics. Therefore, our study may foster the understanding of keeping silent to a medical error in a teamwork setting. In this study, there is no sufficient data to assess whether or not being blamed might impact to students' future decision to speak up. Further study is needed to explore this group to understand if they would have a different perspective.

Limitations

There are some limitations to our study. This is a cross-sectional descriptive study based on one cohort of medical students at single institution in Northern Taiwan. These results may not generalize to other institutions, with different contexts and cultures. Since speaking up behavior in the event of medical error is not a simple communication skill, social factors, cultural factors, interpersonal factors, hospital policies and even rules in individual clinical units may impact students' decisions. We recognize that there may be other dominant factors that could contribute to medical students' response to medical errors. Despite this, we have revealed some important information regarding medical students' thoughts about facing medical error events, this

information could be used for educators and clinical mentors in their educational activities. In addition, while we are encouraged that 100% of the students completed the survey, we recognize their responses could be influenced by the same hesitation for speaking up in the event of a medical error – fear of reprisal. Future work can include follow-up focus groups and interviews to further explore this possibility

Conclusions

Our study findings suggest medical students' perspectives on barriers for speaking up when medical errors occur include characteristics of senior personnel, such as hierarchy, age gap, and negative personality of faculty. Factors, such as potential penalty, workplace atmosphere and acceptance, and teamwork relationships also act as an important role in students' decision to react to medical errors. These results are important for medical educators and clinical mentors for designing personalized educational activities related to speaking up for patient safety.

Disclosure statement

Yi-Chun Chen, S. Barry Issenberg, Zachary Issenberg, Hui-Wen Chen, Yi-No Kang, and Jen-Chieh Wu declare that they have nothing to disclose regarding financial or non-financial conflicts of interest to this manuscript.

Authors contributions

Yi-Chun Chen, Yi-No Kang, and Jen-Chieh Wu: conceptualization. Yi-Chun Chen, Hui-Wen Chen, and Jen-Chieh Wu: Hui-Wen Chen: formal analysis. Yi-Chun Chen, Yi-No Kang, and Jen-Chieh Wu: investigation. Yi-No Kang: methodology. S. Barry Issenberg and Jen-Chieh Wu: supervision. Yi-Chun Chen and Jen-Chieh Wu: writing – original draft. S. Barry Issenberg, Zachary Issenberg, Hui-Wen Chen, and Yi-No Kang: writing – review & editing.

Data availability statement

Data in this study is available to other researchers upon reasonable request.

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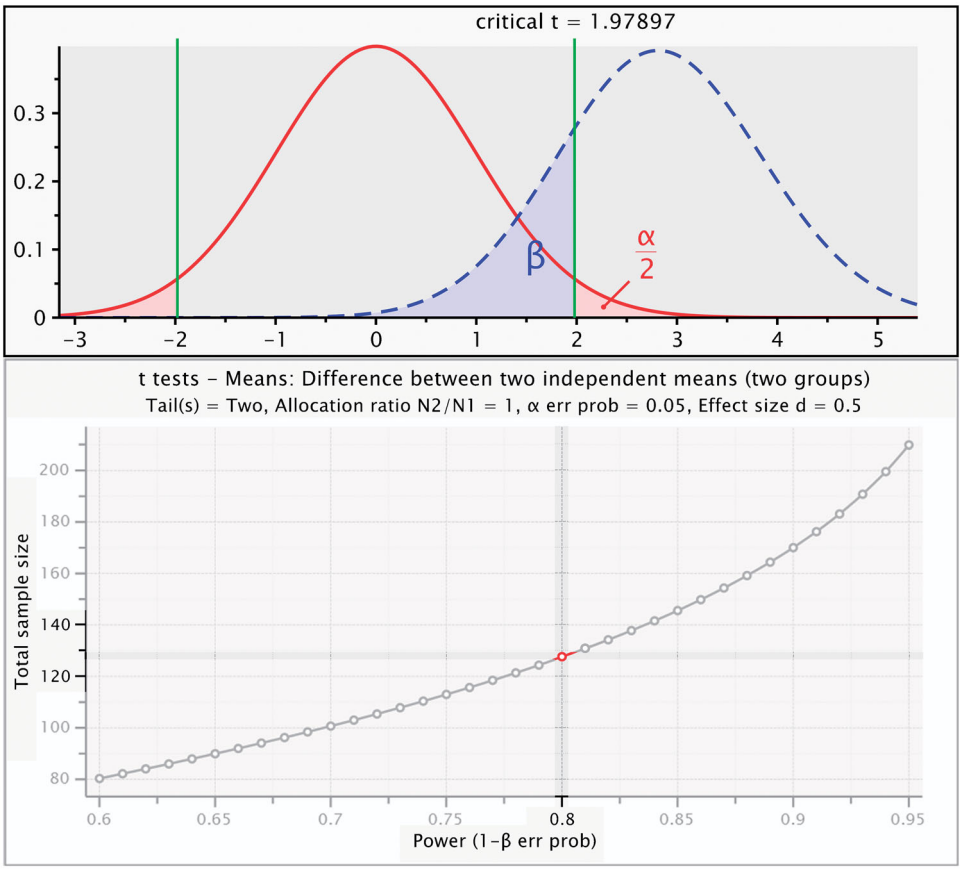
Appendix Figure



Appendix Table. Example of questionnaire

Domain and item		Not important <= = = = = => Very important					
Conflict of interests/social relationship							
01	Penalty/repercussion	①	②	③	④	⑤	
02	Reward or prestige	①	②	③	④	⑤	
03	Social acceptance or rejection	①	②	③	④	⑤	
04	Teamwork/Professionalism	①	②	③	④	⑤	
05	Relationship between senior and me	①	②	③	④	⑤	
Personal (self) Domain							
06	Knowledge/understanding	①	②	③	④	⑤	
07	Communication skills	①	②	③	④	⑤	
Senior Personnel Domain							
08	Hierarchy gap	①	②	③	④	⑤	⑥
09	Age gap	①	②	③	④	⑤	⑥
10	Strict	①	②	③	④	⑤	⑥
11	Grumpy	①	②	③	④	⑤	⑥
12	Stubborn	①	②	③	④	⑤	⑥
13	Male gender	①	②	③	④	⑤	⑥
14	Female gender	①	②	③	④	⑤	⑥
15	Interest conflict	①	②	③	④	⑤	⑥
16	Doctor background	①	②	③	④	⑤	⑥
17	Nurse background	①	②	③	④	⑤	⑥